

Ibn Tofail University

Rigid body mechanics

Problem Set IV

Exercise 1:

A perfect rigid body having the shape of a square $ABCD$ is moved from position $A_1B_1C_1D_1$ to position $A_2B_2C_2D_2$.

1. Show, using a figure, that the motion of this solid can be considered as the sum of a translation and a rotation about a suitably chosen fixed point.
2. Give the value of the angle of rotation.

Correction

Exercise 2:

Let $R(O, X, Y, Z)$ be a reference frame assumed fixed. A homogeneous rod of length L and mass m rotates, in the vertical plane (XOY), about the axis OZ . The position of the rod with respect to R is indicated by the angle θ (See Figure below).

Determine the kinetic elements of the solid, at point O , with respect to frame R .

Correction

Exercise 3:

A homogeneous and non-deformable hoop of radius R , center G , and mass m , rolls without slipping on an inclined plane of angle α .

We denote by x the abscissa of G and by θ the angle that indicates the position of a point on the hoop. Initially, the geometric contact point coincides with point O and $\theta = 0$.

1. What is the number of degrees of freedom of the hoop?
2. Determine the kinetic elements of the hoop at point O .

Correction

Exercise 4:

A homogeneous solid cone S , of mass m , radius R , and height h is in motion about its fixed apex O , relative to a reference frame $T_1(O, X_1, Y_1, Z_1)$. Let $T(O, X, Y, Z)$ be the frame attached to the solid such that the OZ axis is the axis of revolution symmetry for the solid.

1. What is the number of degrees of freedom of the solid?
2. Express, in the Resal basis, the rotation vector of S with respect to T_1 .
3. Determine the kinetic elements of the solid at point O with respect to T_1 .

Correction

Exercise 5:

A homogeneous and non-deformable square plate $OABC$, of side a and mass m , is in motion about point O , origin of a reference frame $T_1(O, X_1, Y_1, Z_1)$, such that one of its sides OA remains in contact with the plane OX_1Y_1 .

1. Represent, on a figure, the frame $T(O, X, Y, Z)$ attached to the solid and identify the Euler angles associated with this solid.
2. What is the number of degrees of freedom of the system?
3. Determine the inertia operator J_O of the solid at point O .
4. Express, in the basis of T , the rotation vector of the solid with respect to T_1 .
5. Express, in the basis of T , the kinetic torsor of the solid, at O , with respect to T_1 .
6. Determine the kinetic energy of the solid with respect to T_1 .

Correction